Aneek Mukherjee

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Work Experience

Standard Bots **Jun. 2025 – Present**

Staff Software Engineer *New York, NY*

- Built a unified C++ robotic gripper control system to consolidate fragmented logic across software layers, reducing integration time for new robotic peripherals and accelerating our AI development.
- Drove hardware scalability vision across software, replacing legacy hard-coded configs with automated runtime detection and dynamic configuration systems — improving safety, reducing support costs, and enabling product-agnostic builds across new form factors with no additional dev effort.
- Spearheaded company-wide agentic engineering best practices; designed custom skills, documented development patterns, and led sessions that accelerated the usage of AI agents in the software org.
- Architected and led the migration to EtherCAT, managing product and leadership stakeholders through rollout; improved data bandwidth and topology flexibility unlocked velocity across both software and hardware orgs.

Amazon Robotics **Aug. 2020 – May 2025**

Software Development Engineer II | Mar, 2022 – May, 2025 *Seattle, WA*

- Spearheaded technical strategy for a 10-developer team — owning decisions on software-controls interfaces, infrastructure, operational excellence, and scalable architectural patterns — to accelerate team velocity.
- Pioneered new embedded firmware capabilities for next-generation autonomous mobile sortation robots, training 4 engineers in C++/Rust while developing robust applications with FreeRTOS, FOC motor control, sensor fusion, wireless communication (BLE, ultra wide band), and BSPs for custom hardware.
- Implemented comprehensive embedded C++ tooling from the ground up that aligned with Amazon-standard patterns, reducing onboarding time and enabling new engineers to effectively contribute to complex firmware projects within 2 weeks with out-of-the-box cross-compilation, unit testing, code organization, and linting.
- Designed and maintained highly scalable, fault-tolerant microservice-based cloud systems orchestrating 4,000+ robots with 99.95% uptime, resulting in 22% YoY operational cost reduction.
- Architected and led delivery of 4 edge-deployed robotic products that automate millions of operations daily in key fulfillment processes in 100+ warehouses (palletization, sortation, mobility), leveraging AWS Outposts & IoT Greengrass to achieve $50M+ annual savings in total.
- Mentored 6 junior engineers and interns through regular 1:1s, code reviews, and technical workshops, fostering team development that resulted in above-bar retention rates and guided 2 engineers to successful mid-level promotions within 2 years.

Software Development Engineer I | Aug, 2020 – Mar, 2022

- Built fleet management, configuration, and provisioning tools for 3 AGV projects, enabling 200+ machine deployments and $13M in annualized labor savings, with excellent operational visibility.

Amazon Robotics **May 2019 – Aug. 2019**

Software Development Engineer Intern *Seattle, WA*

- Engineered a serverless AWS backend and React UI for end-to-end real-time robot event tracking and analysis, processing 100+ TPS using AWS Lambda, Elasticsearch, and Kinesis.

Education

Carnegie Mellon University **May 2020**

Bachelor of Science, Electrical and Computer Engineering *Pittsburgh, PA*

- Teaching Assistant for 15-441/641: Computer Networks, Men's Water Polo Club President

Phillips Exeter Academy **May 2016**

High School *Exeter, NH*

Skills

Languages: C/C++20, Rust, Java, Javascript/Typescript, Python, Go, C#
Embedded: Custom BSP, FreeRTOS, Bare-metal, SPI, I2C, CAN, ARM, STM32, PCB Bring-up
Cloud: AWS (Lambda, DynamoDB, IoT, CloudWatch, SQS, ECS, CDK), Microservices, React, Svelte